

IF330 Pemrograman Web.

02 - Basic PHP Programming

Fenina Adline Twince Tobing

I Putu Ananta Yogiswara

PROGRAM STUDI INFORMATIKA
FAKULTAS TEKNIK DAN INFORMATIKA
UNIVERSITAS MULTIMEDIA NUSANTARA
SEMESTER GANJIL TAHUN AJARAN 2022/2023

REVIEW



- Basic Web Architecture
- Request-Response Cycle
- PHP Introduction
- PHP Hello World
- PHP Variables

TODAY's OUTLINE



- Data Types in PHP
- Strings
- Arrays
- Constants
- Conditional Statement
- Loop Statement
- Function

Data Types in PHP

Scalar

boolean

integer

float

string

Compound

array

object

Special

resource

null

Scalar Data Types

- **Boolean**
 - Represent two possible states: **TRUE** or **FALSE**
- **Integer**
 - Non-decimal number between **-2,147,483,648** and **2,147,483,647**
 - Rules:
 - at least one digit
 - must not have a decimal point
 - can be either positive or negative
 - can be defined in three formats: decimal, hexadecimal, octal
- **Float**
 - A number with a decimal point or a number in exponential form
- **String**

Type Casting in PHP

- **Implicit Type Casting**
 - PHP will automatically convert the type of a variable based on the value assigned to them.
- **Explicit Type Casting**
 - done by programmers using keywords
 - (int) or (integer) - cast to an Integer
 - (bool) or (boolean) - cast to a Boolean
 - (float) or (double) or (real) - cast to a Float
 - (string) - cast to a String
 - (array) - cast to an Array
 - (object) - cast to an Object

Type Casting in PHP

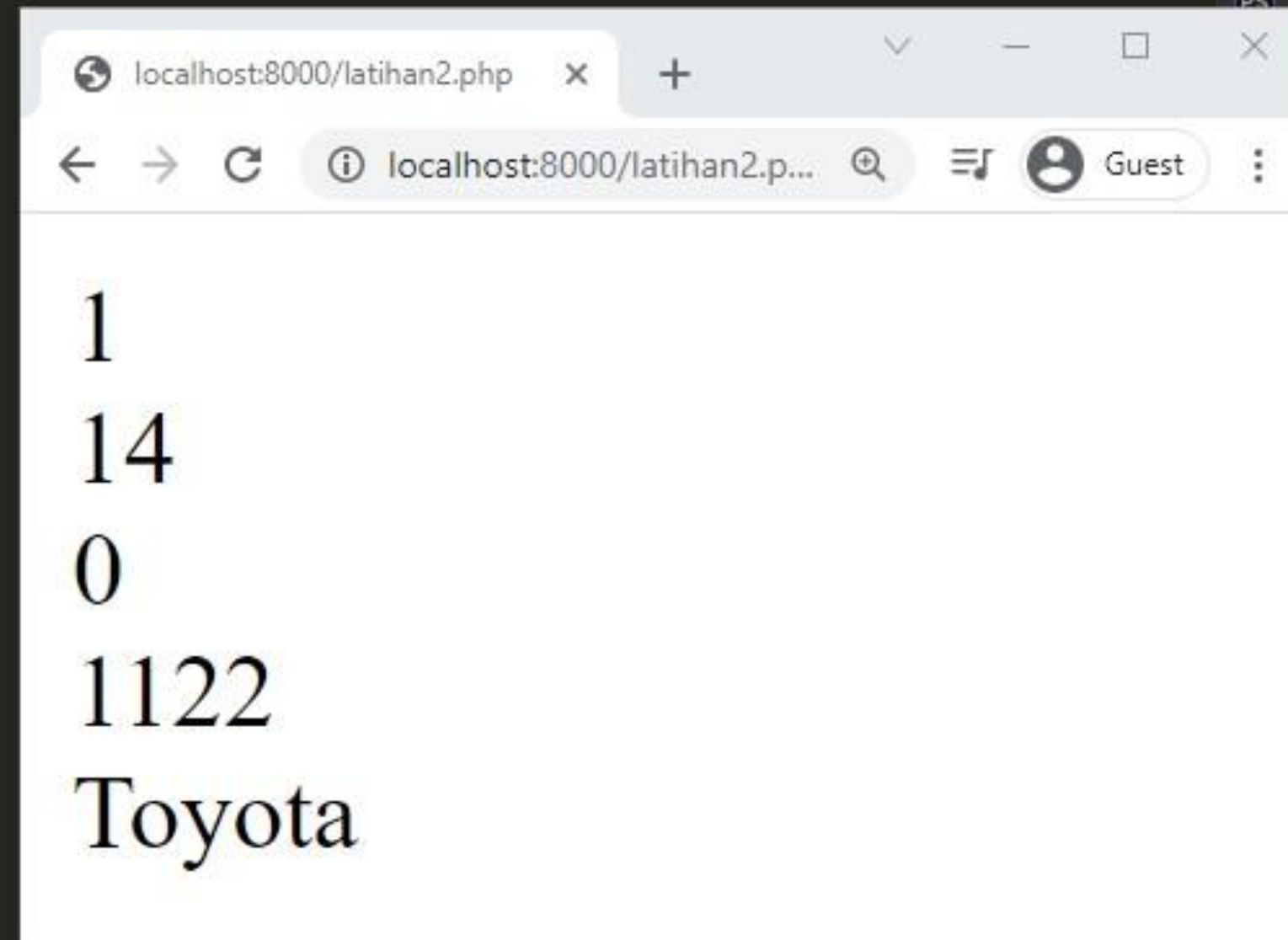
```
<?php
$score = (boolean) 13;
echo $score, "<br />";

$score2 = (int) 14.8;
echo $score2, "<br />";

$sentence = "This is a sentence";
echo (int) $sentence, "<br />";

$score = 1122;
$scoreboard = (array) $score;
echo $scoreboard[0], "<br />";

$model = "Toyota";
$obj = (object) $model;
print $obj->scalar;
?>
```



Strings (1)

A string is a sequence of chars

```
$stringTest = "this is a sequence of chars";  
echo $stringTest[0]; output: t  
echo $stringTest; output: this is a sequence of chars
```

A single quoted strings is displayed "as-is"

```
$age = 37;  
$stringTest = 'I am $age years old'; // output: I am $age years old  
$stringTest = "I am $age years old"; // output: I am 37 years old
```

Concatenation

```
$conc = "is" . "a" . "composed" . "string"  
echo $conc; // output: is a composed string  
$newConc = 'Also $conc ' . $conc;  
echo $newConc; // output: Also $conc is a composed string
```


Strings (2)

Explode Function

```
$sequence = "A, B, C, D, E, F, G";  
$elements = explode(",", $sequence);  
// Now elements in an array all substrings between  
// "," char  
echo $elements[0]; // output: A  
echo $elements[1]; // output: B  
echo $elements[2]; // output: C;  
echo $elements[3]; // output: D  
echo $elements[4]; // output: E  
echo $elements[5]; // output: F  
echo $elements[6]; // output: G
```

Arrays: Indexed

Indexed Arrays

```
$students = array("John Thor", "John Wick", "John Doe");
```

OR

```
$students[0] = "John Thor";
```

```
$students[1] = "John Wick";
```

```
$students[2] = "John Doe";
```

```
// Loop through an Indexed Arrays  
for($i=0; $i<$arrLength; $i++){  
    echo $students[$i];  
}  
  
// OR  
foreach($students as $student){  
    echo $student;  
}
```

Arrays: Associative

Associative Arrays

```
$age = array("Abang" => 23, "John" => 34, "Thor" => 45);
```

OR

```
$age["Abang"] = 23;
```

```
$age["John"] = 34;
```

```
$age["Thor"] = 45;
```

```
// Loop through an Associative Arrays
$age = array("Abang" => 23, "John" => 34, "Thor" => 45);
foreach($age as $x => $x_value){
    echo "Key: " . $x . ", Value: " . $x_value;
    echo "<br />";
}
```

Arrays: Multidimensional

Multidimensional Arrays

```
$cars = array (  
    array("Volvo", 22, 18) ,  
    array("BMW", 15, 13) ,  
    array("Saab", 5, 2) ,  
    array("Land  
Rover", 17, 15)  
);
```

Name	Stock	Sold
Volvo	22	18
BMW	15	13
Saab	5	2
Land Rover	17	15

Arrays: Sorting

Sort Functions for Arrays

- **sort()** - sort arrays in ascending order
- **rsort()** - sort arrays in descending order
- **asort()** - sort associative arrays in ascending order, according to the value
- **ksort()** - sort associative arrays in ascending order, according to the key
- **arsort()** - sort associative arrays in descending order, according to the value
- **krsort()** - sort associative arrays in descending order, according to the key

Constants

Syntax

```
define(name, value)
```

```
define("GREETING", "welcome to UMN!");  
echo GREETING;
```

// Constants are automatically global and can be used across the entire script

Conditional Statements

- **if** statement - executes some code if one condition is true
- **if...else** statement - executes some code if condition is true and another code if that condition is false
- **if...elseif...else** statement - executes different codes for more than two conditions
- **switch** statement - selects one of many blocks of code to be executed

Loop Statements

- **while** - loops through a block of code as long as the specified condition is true
- **do...while** - loops through a block of code once, and then repeats the loop as long as the specified condition is true
- **for** - loops through a block of code a specified number of times
- **foreach** - loops through a block of code for each element in an array

Functions

Syntax

```
function functionName([parameters-list] opt) {  
    ...code to be executed...  
}
```

- It's not allowed to overload the name of an existing function
- Function names aren't case-sensitive
- To each parameter can be assigned a default value
- Arguments can be passed by value or by reference
- It's possible using a variable number of parameters

Functions

// Example 1

```
function hello(){  
    echo "Hello world!";  
}  
hello();
```

// Example 2

```
function hello(){  
    return "Hello world!";  
}  
echo hello();
```

// Example 3

```
function hello($name, int $pos){  
    return "Hello $name, your  
position is $pos";  
}
```

```
echo hello("John", 5);  
echo hello("Thor");  
echo hello("Doe", "9");
```

SUMMARY

After finishing this module, you should be able:

- to understand various data types in PHP
- to understand what is type casting in PHP
- to understand strings & arrays operations
- to understand constants in PHP
- to understand Conditional and Loop Statement
- to understand Functions and variable scope

NEXT WEEK'S OUTLINE

- Pre-defined variables, constants and functions
- PHP Forms



next week

The text "next week" is written in a bold, yellow, cursive font with a thick blue outline. A long, thick blue underline is positioned beneath the text, starting from the beginning of the word "next" and extending past the end of "week".

REFERENCES

- <https://www.w3schools.com/PHP/default.asp>
- <https://www.php.net/docs.php>
- Luke Welling & Laura Thomson. 2017. PHP and MySQL Web Development (5th ed)
- Robin Nixon. 2021. Learning PHp, MySQL & JavaScript.

Visi

Menjadi Program Studi Strata Satu Informatika **unggulan** yang menghasilkan lulusan **berwawasan internasional** yang **kompeten** di bidang Ilmu Komputer (*Computer Science*), **berjiwa wirausaha** dan **berbudi pekerti luhur**.



Misi

1. Menyelenggarakan pembelajaran dengan teknologi dan kurikulum terbaik serta didukung tenaga pengajar profesional.
2. Melaksanakan kegiatan penelitian di bidang Informatika untuk memajukan ilmu dan teknologi Informatika.
3. Melaksanakan kegiatan pengabdian kepada masyarakat berbasis ilmu dan teknologi Informatika dalam rangka mengamalkan ilmu dan teknologi Informatika.